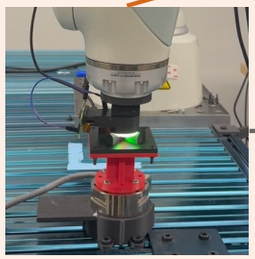
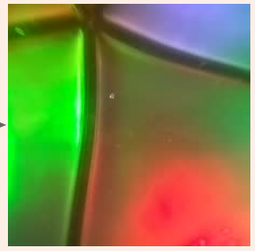
(a) Register real contacts

Labels need calibration and verification

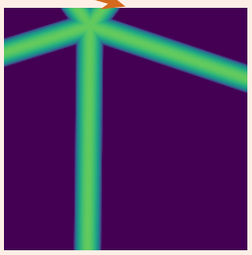
q^* from the robot pose



Registered press

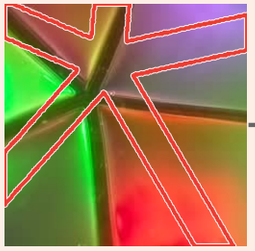


Real image



GT label

after re-placing the object



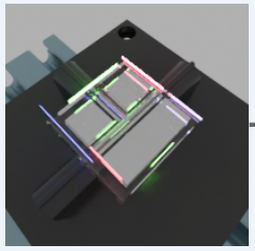
old calibration



re-calibrated, verified

(b) Simulate contacts

Synthetic RGB needs sensor-response modeling

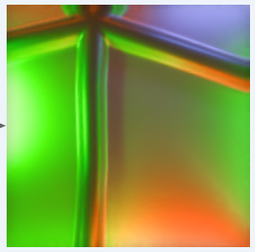


Simulated contact

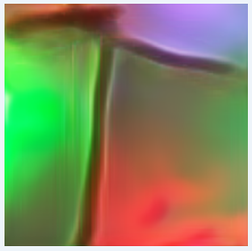


GT label (free)

render

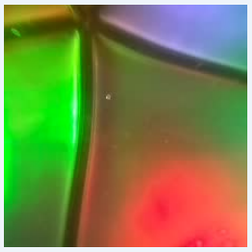


PBR



Learned
(fitted on real pairs)

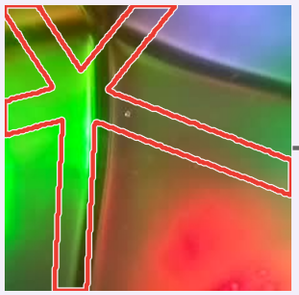
$\neq$



Real image

(c) AlignTac: real appearance + aligned geometry

unlabeled real image + known mesh



mesh at teacher estimate

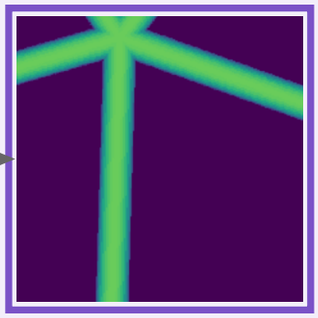
Learned aligner $\mathcal{A}$

train

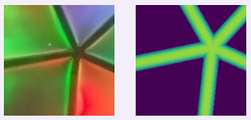
$q \leftarrow q + \Delta q$



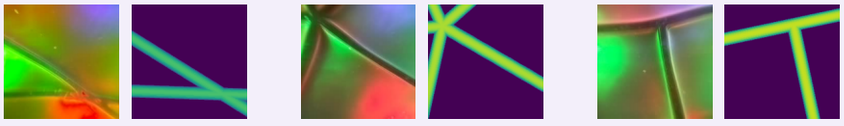
mesh after alignment



Pseudo-label



K **registered contacts**



pseudo-labels for the remaining real images

K=25 with AlignTac ≈
real-only K=50
(contact metrics)